

	$\mathcal{K}_{0^+}^{\#1} \dagger$	$\mathcal{K}_{0^+}^{\#1}$	$\alpha\beta\chi$			
	$\mathcal{K}_{0^+}^{\#1} \dagger$	0	0			
$\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0				
		$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha}$	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha}$	
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\Upsilon_1 k^2$	$-\frac{\Upsilon_2 k^2}{2\sqrt{2}}$	0	0	
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$-\frac{\Upsilon_2 k^2}{2\sqrt{2}}$	$\frac{\Upsilon_3 k^2}{2}$	0	0	
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha}$	0	0	$\frac{\Upsilon_4 k^2}{2}$	$\frac{\Upsilon_4 k^2}{\sqrt{2}}$	
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha}$	0	0	$\frac{\Upsilon_4 k^2}{\sqrt{2}}$	$\Upsilon_4 k^2$	
				$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	
				$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0
				$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{3k^2}{2} \kappa_{15}^{(4)}$

	$\mathcal{T}_{0^+}^{\#1} \dagger$	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^+}^{\#1 \alpha\beta\chi}$			
	$\mathcal{T}_{0^+}^{\#1} \dagger$	0	0			
	$\mathcal{T}_{0^+}^{\#1 \alpha\beta\chi}$	0	0			
		$\mathcal{T}_{1^+}^{\#1} \alpha\beta$	$\mathcal{T}_{1^+}^{\#2} \alpha\beta$	$\mathcal{T}_{1^+}^{\#1} \alpha$	$\mathcal{T}_{1^+}^{\#2} \alpha$	
	$\mathcal{T}_{1^+}^{\#1} \dagger \alpha\beta$	$\frac{\Upsilon_3 k^2}{2 \text{Det}(1^+)}$	$\frac{\Upsilon_2 k^2}{2 \sqrt{2} \text{Det}(1^+)}$	0	0	
	$\mathcal{T}_{1^+}^{\#2} \dagger \alpha\beta$	$\frac{\Upsilon_2 k^2}{2 \sqrt{2} \text{Det}(1^+)}$	$\frac{\Upsilon_1 k^2}{\text{Det}(1^+)}$	0	0	
	$\mathcal{T}_{1^+}^{\#1} \dagger \alpha$	0	0	$\frac{1}{3 \text{Det}(1^+)}$	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	
	$\mathcal{T}_{1^+}^{\#2} \dagger \alpha$	0	0	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{2}{3 \text{Det}(1^+)}$	
				$\mathcal{T}_{2^+}^{\#1} \alpha\beta$	$\mathcal{T}_{2^+}^{\#1} \alpha\beta\chi$	
				$\mathcal{T}_{2^+}^{\#1} \dagger \alpha\beta$	0	0
				$\mathcal{T}_{2^+}^{\#1} \dagger \alpha\beta\chi$	0	$\frac{2}{3 k^2 \kappa_{15}^{(4)}}$

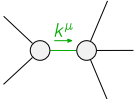
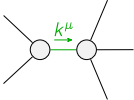
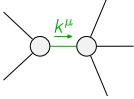
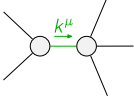
Abbreviations used in matrices

$$\Upsilon_1 == 2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)} \ \&\& \ \Upsilon_2 == 2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_4^{(4)} \ \&\& \ \Upsilon_3 == 9 \mathcal{K}_1^{(4)} + 11 \mathcal{K}_{15}^{(4)} + \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)} \ \&\& \ \Upsilon_4 == 2 \mathcal{K}_1^{(4)} + 3 \mathcal{K}_{15}^{(4)} \ \&\&$$

$$\text{Det}(1^+) == \frac{1}{8} k^4 (84 \mathcal{K}_{15}^{(4)} + 36 \mathcal{K}_1^{(4)} (2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)}) + 12 \mathcal{K}_{15}^{(4)} (-3 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)}) - (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2)$$

Lagrangian

$$\begin{aligned} &\mathcal{K}_1 \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^\chi \mathcal{K}_{\alpha}^\alpha \beta - \mathcal{K}_1 \partial_\chi \mathcal{K}_{\beta\delta}^\delta \partial^\chi \mathcal{K}_{\alpha}^\alpha \beta + \mathcal{K}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \\ &\mathcal{K}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + 3 \mathcal{K}_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \mathcal{K}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + 6 \mathcal{K}_1 \partial^\chi \mathcal{K}_{\alpha}^\alpha \beta \partial_\delta \mathcal{K}_{\chi\beta}^\delta + \\ &6 \mathcal{K}_{15} \partial^\chi \mathcal{K}_{\alpha}^\alpha \beta \partial_\delta \mathcal{K}_{\chi\beta}^\delta + \frac{9}{2} \mathcal{K}_1 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{9}{2} \mathcal{K}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \\ &\frac{1}{2} \mathcal{K}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{1}{2} \mathcal{K}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \mathcal{K}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \mathcal{K}_{15} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{J}^\alpha{}_\alpha{}^\beta == 0$	
$\mathcal{J}_{1^+}^{\#1\alpha} == \mathcal{J}_{1^+}^{\#2\alpha}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta == 0$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^{\chi}{}_\chi{}^\delta == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^{\chi}{}_\chi{}^\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	10		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\begin{aligned} &-((-144 \mathcal{K}_1^{(4)2} \mathcal{K}_{15}^{(4)} - 408 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)2} - 288 \mathcal{K}_{15}^{(4)3} + 72 \mathcal{K}_1^{(4)2} \mathcal{K}_3^{(4)} + \\ &196 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} + 132 \mathcal{K}_{15}^{(4)2} \mathcal{K}_3^{(4)} + 8 \mathcal{K}_1^{(4)} \mathcal{K}_3^{(4)2} + 12 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)2} - 28 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} \mathcal{K}_4^{(4)} - 42 \mathcal{K}_{15}^{(4)2} \mathcal{K}_4^{(4)} + \\ &8 \mathcal{K}_1^{(4)} \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} + 12 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} + 2 \mathcal{K}_1^{(4)} \mathcal{K}_4^{(4)2} + 3 \mathcal{K}_{15}^{(4)} \mathcal{K}_4^{(4)2} + \sqrt{((2 \mathcal{K}_1^{(4)} + 3 \mathcal{K}_{15}^{(4)})^2} \\ &(20196 \mathcal{K}_{15}^{(4)4} + 324 \mathcal{K}_1^{(4)2} (49 \mathcal{K}_{15}^{(4)2} - 48 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} + 12 \mathcal{K}_3^{(4)2}) - 24 \mathcal{K}_{15}^{(4)3} (701 \mathcal{K}_3^{(4)} - 242 \mathcal{K}_4^{(4)} + \\ &4 \mathcal{K}_{15}^{(4)} (50 \mathcal{K}_3^{(4)} - 17 \mathcal{K}_4^{(4)}) (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + 3 (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^4 + 12 \mathcal{K}_{15}^{(4)2} (143 \mathcal{K}_3^{(4)2} - 360 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 5 \mathcal{K}_4^{(4)2}) + \\ &72 \mathcal{K}_1^{(4)} (489 \mathcal{K}_{15}^{(4)3} + 3 \mathcal{K}_3^{(4)} (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + \mathcal{K}_{15}^{(4)2} (-439 \mathcal{K}_3^{(4)} + 70 \mathcal{K}_4^{(4)}) + \mathcal{K}_{15}^{(4)} (76 \mathcal{K}_3^{(4)2} - 58 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 6 \mathcal{K}_4^{(4)2})))))/ \\ &(\mathcal{K}_{15}^{(4)} (2 \mathcal{K}_1^{(4)} + 3 \mathcal{K}_{15}^{(4)}) (84 \mathcal{K}_{15}^{(4)2} + 36 \mathcal{K}_1^{(4)} (2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)}) + 12 \mathcal{K}_{15}^{(4)} (-3 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)}) - (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2))) \end{aligned}$
	1	0	$\begin{aligned} &(144 \mathcal{K}_1^{(4)2} \mathcal{K}_{15}^{(4)} + 408 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)2} + 288 \mathcal{K}_{15}^{(4)3} - 72 \mathcal{K}_1^{(4)2} \mathcal{K}_3^{(4)} - \\ &196 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} - 132 \mathcal{K}_{15}^{(4)2} \mathcal{K}_3^{(4)} - 8 \mathcal{K}_1^{(4)} \mathcal{K}_3^{(4)2} - 12 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)2} + 28 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} \mathcal{K}_4^{(4)} + \\ &42 \mathcal{K}_{15}^{(4)2} \mathcal{K}_4^{(4)} - 8 \mathcal{K}_1^{(4)} \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 12 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 2 \mathcal{K}_1^{(4)} \mathcal{K}_4^{(4)2} - 3 \mathcal{K}_{15}^{(4)} \mathcal{K}_4^{(4)2} + \sqrt{((2 \mathcal{K}_1^{(4)} + 3 \mathcal{K}_{15}^{(4)})^2} \\ &(20196 \mathcal{K}_{15}^{(4)4} + 324 \mathcal{K}_1^{(4)2} (49 \mathcal{K}_{15}^{(4)2} - 48 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} + 12 \mathcal{K}_3^{(4)2}) - 24 \mathcal{K}_{15}^{(4)3} (701 \mathcal{K}_3^{(4)} - 242 \mathcal{K}_4^{(4)} + \\ &4 \mathcal{K}_{15}^{(4)} (50 \mathcal{K}_3^{(4)} - 17 \mathcal{K}_4^{(4)}) (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + 3 (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^4 + 12 \mathcal{K}_{15}^{(4)2} (143 \mathcal{K}_3^{(4)2} - 360 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 5 \mathcal{K}_4^{(4)2}) + \\ &72 \mathcal{K}_1^{(4)} (489 \mathcal{K}_{15}^{(4)3} + 3 \mathcal{K}_3^{(4)} (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + \mathcal{K}_{15}^{(4)2} (-439 \mathcal{K}_3^{(4)} + 70 \mathcal{K}_4^{(4)}) + \mathcal{K}_{15}^{(4)} (76 \mathcal{K}_3^{(4)2} - 58 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 6 \mathcal{K}_4^{(4)2})))))/ \\ &(\mathcal{K}_{15}^{(4)} (2 \mathcal{K}_1^{(4)} + 3 \mathcal{K}_{15}^{(4)}) (84 \mathcal{K}_{15}^{(4)2} + 36 \mathcal{K}_1^{(4)} (2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)}) + 12 \mathcal{K}_{15}^{(4)} (-3 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)}) - (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2))) \end{aligned}$
	2	0	$\begin{aligned} &-((216 \mathcal{K}_{15}^{(4)3} + 36 \mathcal{K}_1^{(4)2} (8 \mathcal{K}_{15}^{(4)} - 7 \mathcal{K}_3^{(4)}) - 216 \mathcal{K}_{15}^{(4)2} \mathcal{K}_3^{(4)} - 60 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)2} + 108 \mathcal{K}_{15}^{(4)2} \mathcal{K}_4^{(4)} - \\ &60 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 15 \mathcal{K}_{15}^{(4)} \mathcal{K}_4^{(4)2} + \mathcal{K}_1^{(4)} (648 \mathcal{K}_{15}^{(4)2} + 36 \mathcal{K}_{15}^{(4)} (-16 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)}) - 7 (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2) + \\ &\sqrt{(1296 \mathcal{K}_1^{(4)4} (352 \mathcal{K}_{15}^{(4)2} - 256 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} + 49 \mathcal{K}_3^{(4)2}) + 72 \mathcal{K}_1^{(4)3} \\ &(29376 \mathcal{K}_{15}^{(4)3} - 216 \mathcal{K}_{15}^{(4)2} (91 \mathcal{K}_3^{(4)} - 8 \mathcal{K}_4^{(4)}) + 49 \mathcal{K}_3^{(4)} (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + 4 \mathcal{K}_{15}^{(4)} (736 \mathcal{K}_3^{(4)2} - 263 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 32 \mathcal{K}_4^{(4)2})) + \\ &\mathcal{K}_1^{(4)2} (3685824 \mathcal{K}_{15}^{(4)4} - 1728 \mathcal{K}_{15}^{(4)3} (1285 \mathcal{K}_3^{(4)} - 253 \mathcal{K}_4^{(4)}) + 1080 \mathcal{K}_{15}^{(4)} (11 \mathcal{K}_3^{(4)} - \mathcal{K}_4^{(4)}) (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + \\ &49 (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^4 + 864 \mathcal{K}_{15}^{(4)2} (248 \mathcal{K}_3^{(4)2} - 297 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 26 \mathcal{K}_4^{(4)2})) + 6 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} \\ &(485568 \mathcal{K}_{15}^{(4)4} - 1728 \mathcal{K}_{15}^{(4)3} (151 \mathcal{K}_3^{(4)} - 48 \mathcal{K}_4^{(4)}) + 144 \mathcal{K}_{15}^{(4)} (14 \mathcal{K}_3^{(4)} - 3 \mathcal{K}_4^{(4)}) (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + 23 (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^4 + \\ &432 \mathcal{K}_{15}^{(4)2} (24 \mathcal{K}_3^{(4)2} - 106 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 5 \mathcal{K}_4^{(4)2})) + 9 \mathcal{K}_{15}^{(4)2} (101952 \mathcal{K}_{15}^{(4)4} - 1728 \mathcal{K}_{15}^{(4)3} (30 \mathcal{K}_3^{(4)} - 11 \mathcal{K}_4^{(4)}) + \\ &72 \mathcal{K}_{15}^{(4)} (4 \mathcal{K}_3^{(4)} - 3 \mathcal{K}_4^{(4)}) (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + 13 (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^4 + 432 \mathcal{K}_{15}^{(4)2} (4 \mathcal{K}_3^{(4)2} - 20 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} + \mathcal{K}_4^{(4)2})))))/ \\ &(\mathcal{K}_{15}^{(4)} (2 \mathcal{K}_1^{(4)} + 3 \mathcal{K}_{15}^{(4)}) (84 \mathcal{K}_{15}^{(4)2} + 36 \mathcal{K}_1^{(4)} (2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)}) + 12 \mathcal{K}_{15}^{(4)} (-3 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)}) - (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2))) \end{aligned}$
	2	0	$\begin{aligned} &(-216 \mathcal{K}_{15}^{(4)3} - 36 \mathcal{K}_1^{(4)2} (8 \mathcal{K}_{15}^{(4)} - 7 \mathcal{K}_3^{(4)}) + 216 \mathcal{K}_{15}^{(4)2} \mathcal{K}_3^{(4)} + 60 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)2} - 108 \mathcal{K}_{15}^{(4)2} \mathcal{K}_4^{(4)} + \\ &60 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} + 15 \mathcal{K}_{15}^{(4)} \mathcal{K}_4^{(4)2} + \mathcal{K}_1^{(4)} (-648 \mathcal{K}_{15}^{(4)2} + 36 \mathcal{K}_{15}^{(4)} (16 \mathcal{K}_3^{(4)} - \mathcal{K}_4^{(4)}) + 7 (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2) + \\ &\sqrt{(1296 \mathcal{K}_1^{(4)4} (352 \mathcal{K}_{15}^{(4)2} - 256 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} + 49 \mathcal{K}_3^{(4)2}) + 72 \mathcal{K}_1^{(4)3} \\ &(29376 \mathcal{K}_{15}^{(4)3} - 216 \mathcal{K}_{15}^{(4)2} (91 \mathcal{K}_3^{(4)} - 8 \mathcal{K}_4^{(4)}) + 49 \mathcal{K}_3^{(4)} (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + 4 \mathcal{K}_{15}^{(4)} (736 \mathcal{K}_3^{(4)2} - 263 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 32 \mathcal{K}_4^{(4)2})) + \\ &\mathcal{K}_1^{(4)2} (3685824 \mathcal{K}_{15}^{(4)4} - 1728 \mathcal{K}_{15}^{(4)3} (1285 \mathcal{K}_3^{(4)} - 253 \mathcal{K}_4^{(4)}) + 1080 \mathcal{K}_{15}^{(4)} (11 \mathcal{K}_3^{(4)} - \mathcal{K}_4^{(4)}) (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + \\ &49 (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^4 + 864 \mathcal{K}_{15}^{(4)2} (248 \mathcal{K}_3^{(4)2} - 297 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 26 \mathcal{K}_4^{(4)2})) + \\ &6 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} (485568 \mathcal{K}_{15}^{(4)4} - 1728 \mathcal{K}_{15}^{(4)3} (151 \mathcal{K}_3^{(4)} - 48 \mathcal{K}_4^{(4)}) + 144 \mathcal{K}_{15}^{(4)} (14 \mathcal{K}_3^{(4)} - 3 \mathcal{K}_4^{(4)}) (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + \\ &23 (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^4 + 432 \mathcal{K}_{15}^{(4)2} (24 \mathcal{K}_3^{(4)2} - 106 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} - 5 \mathcal{K}_4^{(4)2})) + \\ &9 \mathcal{K}_{15}^{(4)2} (101952 \mathcal{K}_{15}^{(4)4} - 1728 \mathcal{K}_{15}^{(4)3} (30 \mathcal{K}_3^{(4)} - 11 \mathcal{K}_4^{(4)}) + 72 \mathcal{K}_{15}^{(4)} (4 \mathcal{K}_3^{(4)} - 3 \mathcal{K}_4^{(4)}) (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2 + \\ &13 (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^4 + 432 \mathcal{K}_{15}^{(4)2} (4 \mathcal{K}_3^{(4)2} - 20 \mathcal{K}_3^{(4)} \mathcal{K}_4^{(4)} + \mathcal{K}_4^{(4)2})))))/ \\ &(\mathcal{K}_{15}^{(4)} (2 \mathcal{K}_1^{(4)} + 3 \mathcal{K}_{15}^{(4)}) (84 \mathcal{K}_{15}^{(4)2} + 36 \mathcal{K}_1^{(4)} (2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)}) + 12 \mathcal{K}_{15}^{(4)} (-3 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)}) - (2 \mathcal{K}_3^{(4)} + \mathcal{K}_4^{(4)})^2))) \end{aligned}$